☆ Double Sideband Suppressed Carrier (DSB-SC) system

→ The equation of AM wave in its simplest form i.e. single-tone modulation is expressed as,

$$s(t) = A\cos\omega_c t + A.\frac{m_a}{2}\cos(\omega_c+\omega_m)t + A.\frac{m_a}{2}\cos(\omega_c-\omega_m)t$$

*→ From the above equation, it is obvious that the carrier component in AM wave remains constant in amplitude and frequency.

→ This means that the carrier of AM (Amplitude modulated) wave does not convey any information.

→ In power calculation of AM sig, it has been observed that for single-tone sinusoidal modulation, the ratio of the total power to the carrier power is $\left(1+\frac{m_a^2}{2}\right)$, m_a being the modulation index.

→ Thus for 100% modulation about 67% of the total power is required for transmitting the carrier which does not contain any information.

→ Hence, if the carrier is suppressed, only the sidebands remain and in this way a saving of two-third power may be achieved at 100% modulation.

→ This type of suppression of carrier does not affect the baseband signal in any way.

→ The resulting signal obtained by suppressing the carrier from the modulated wave is called Double sideband suppressed carrier (DSB-SC) system.

*→ Thus, in a Double-sideband suppressed carrier modulation, there is no carrier signal only sidebands are present.

→ We know that the frequency-shifting property of fourier transform is given as under

If, $x(t) \longleftrightarrow X(\omega)$

then $e^{j\omega_c t} x(t) \longleftrightarrow X(\omega-\omega_c)$ —— (1)

This property states that if a signal $x(t)$ is multiplied by $e^{j\omega_c t}$ in time-domain then its spectrum $X(\omega)$ is frequency domain is shifted by an amount ω_c.

Similarly, $e^{-j\omega_c t} x(t) \longleftrightarrow X(\omega+\omega_c)$ —— (2)

But, since $e^{j\omega_c t}$ is not a real function and cannot be generated practically, therefore, frequency shifting in practice is achieved by multiplying $x(t)$ by a sinusoid such as $\cos \omega_c t$

Hence, $x(t) \cos \omega_c t = x(t) \frac{1}{2} (e^{j\omega_c t} + e^{-j\omega_c t})$

$$x(t) \cos \omega_c t = \frac{1}{2} x(t) e^{j\omega_c t} + \frac{1}{2} x(t) e^{-j\omega_c t} \quad \text{—— (3)}$$

Using frequency-shifting property in equation (3), we get,

$$x(t) \cos \omega_c t \longleftrightarrow \frac{1}{2} [X(\omega-\omega_c) + X(\omega+\omega_c)] \quad \text{—— (4)}$$

-> This means that the multiplication of a signal $x(t)$ by a sinusoid of frequency ω_c shifts the spectrum $X(\omega)$ by $\pm\omega_c$.

-> Now, if $x(t)$ is taken as modulating or base band signal and $\cos \omega_c t$ is taken as carrier s/g, then $x(t) \cos \omega_c t$ represents the modulated signal.

-> eq (4) represents spectrum of modulated signal contains only shifted spectrum of signal and there is no carrier components.

" The modulated signal, which contains no carrier but two sidebands is called Double-sideband Suppressed Carrier (DSB-SC) modulation."

-> Hence, the term $x(t) \cos \omega_c t$ represents a DSB-SC signal.

Block Diagram

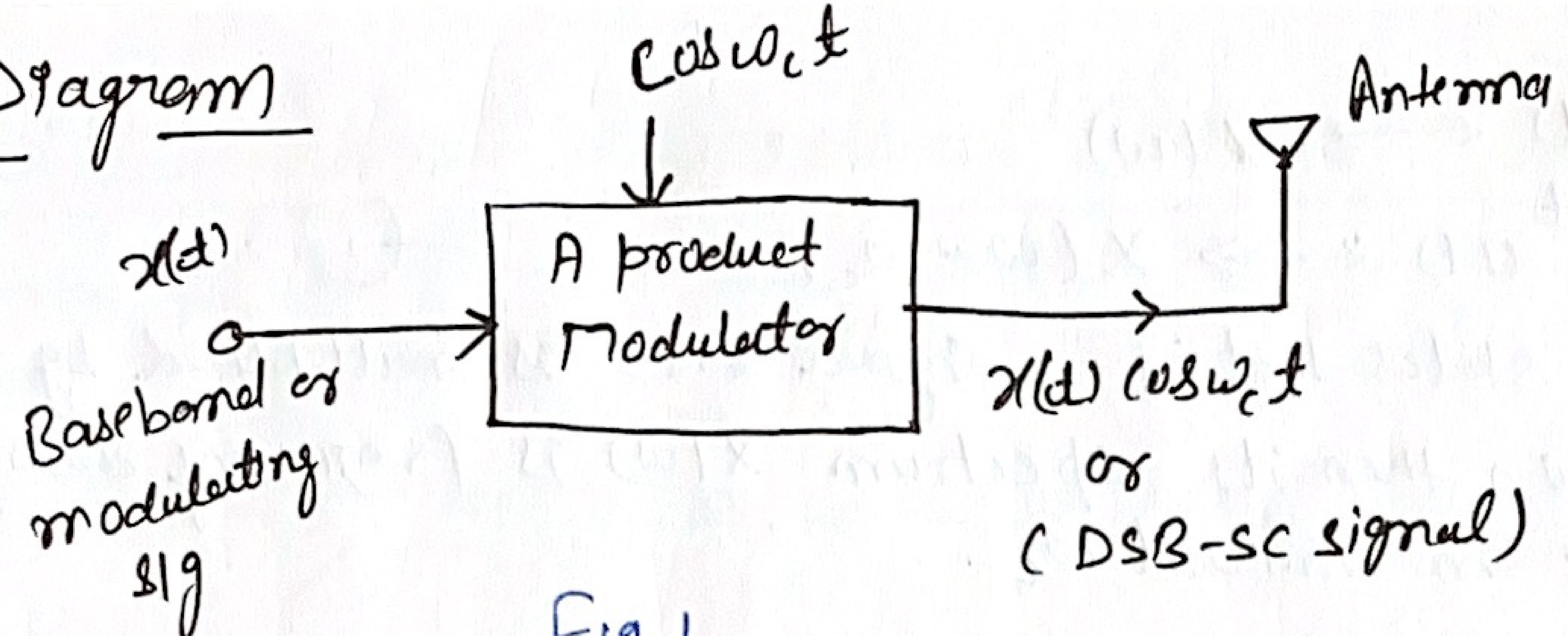


Fig 1

→ DSB-SC signal is obtained by simply multiplying modulating signal $x(t)$ with carrier signal $\cos\omega_c t$. This is achieved by a product modulator

→ Fig 1 shows Block diagram of a product modulator.

→ Fig 2 shows the modulating sig and its Fourier transform, carrier signal and its Fourier transform and DSB-SC signal and its frequency spectrum.

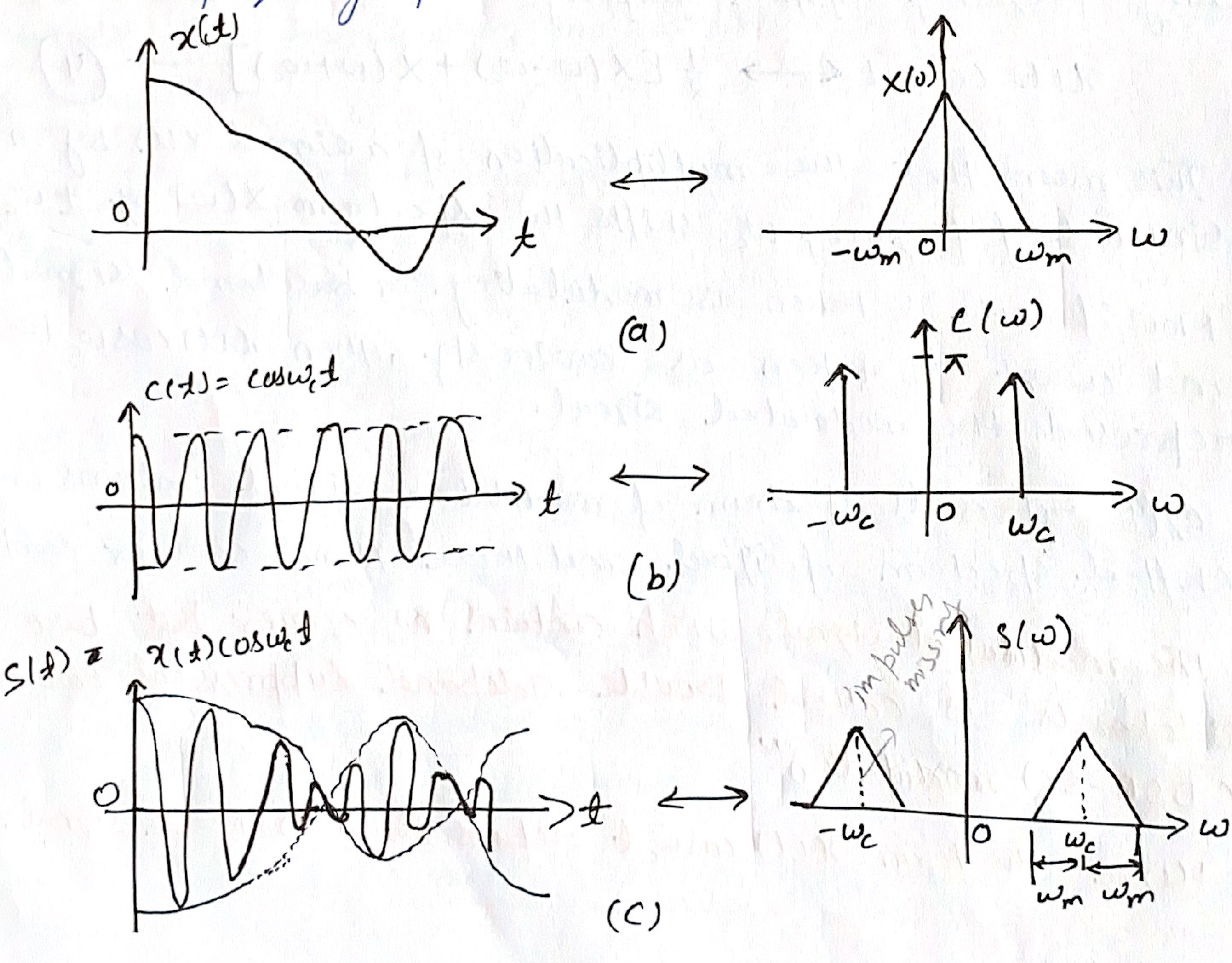


Fig 2

Few important points

1. It is obvious from fig 2, that the DSB-SC signal exhibits phase-reversal at zero crossings i.e. whenever the modulating s/g, $x(t)$ crosses zero. Becoz of this, the evelope of DSB-SC modulated s/g is different from the message signal. This is unlike the case of an AM wave

2. From fig.2., it is also clear that impulses at $\pm \omega_c$ are missing which means that the carrier term is suppressed in the spectrum only two side band terms, USB and LSB are left. Therefore, it is called DSB-SC system.

3. Considering only +ve side, USB $\rightarrow (\omega_c + \omega_B)$, LSB $\rightarrow (\omega_c + \omega_B)$
The difference of these two is equal to the transmission B.W of a DSB-SC s/g. i.e.

$$B.W, B = (\omega_c + \omega_m) - (\omega_c - \omega_m) = 2\omega_m$$

It is obvious that the B.W of DSB-SC modulation is same as that of general AM wave.

☆ Generation of DSB-SC signal :-

→ We know that, to generate the DSB-SC wave, we need to take the product of $x(t)$ and carrier $c(t)$; for this we use product modulator.

→ There are two forms of product modulator.
1) Linear modulator
2) Non-linear modulator

I Linear Modulator for DSB-SC Generation :-

→ "Linear modulator is a system whose gain or Transfer function can be varied with time by applying a time varying signal at certain point".

→ The gain is proportional to signal $f(t)$

$$\boxed{G = K f(t)}$$

G = gain
K = constant of proportionality
$f(t)$ = gain varing signal

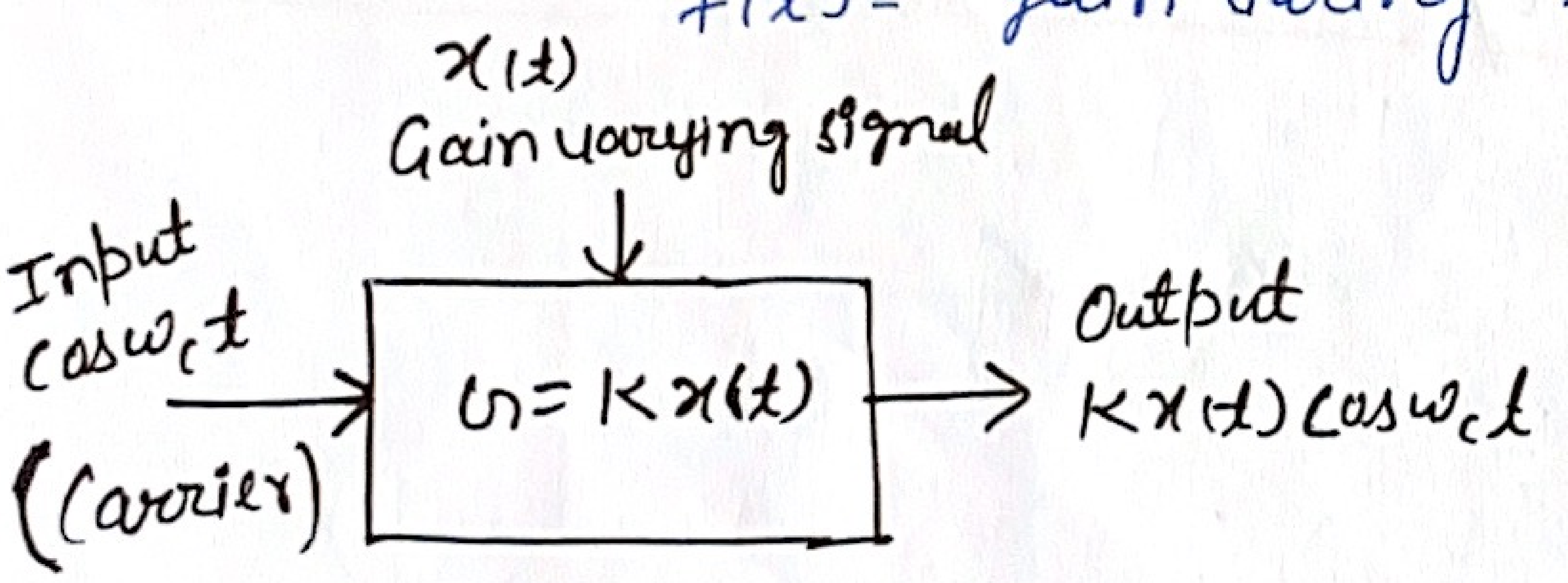


Fig. Linear modular

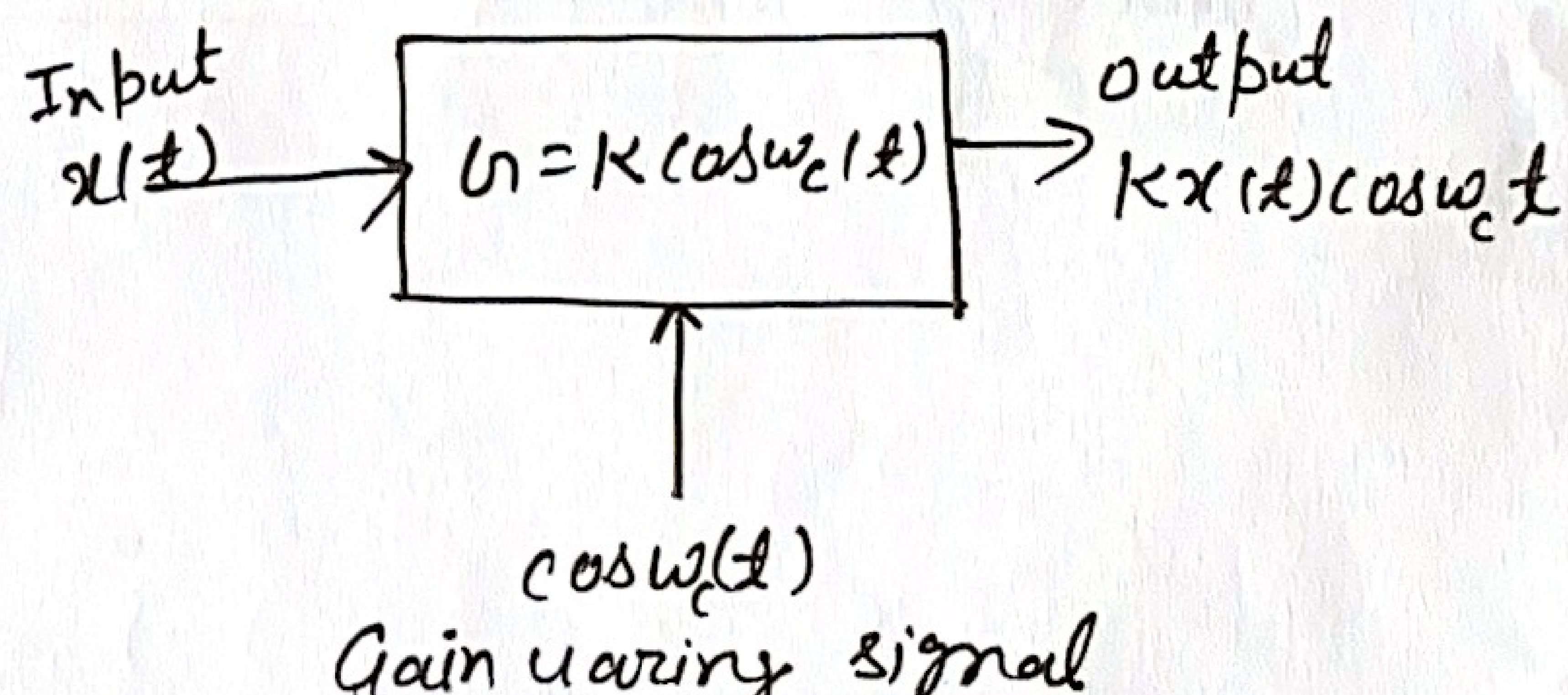


Fig. Another version of linear modulator

Modulator Gain, $G = K x(t)$

O/P ⇒ $V_o = G \times \text{input}$

$$\boxed{V_o = K x(t) \cos\omega_c t}$$

This is a modulated s/g DSB-SC

$G = K \cos\omega_c t$

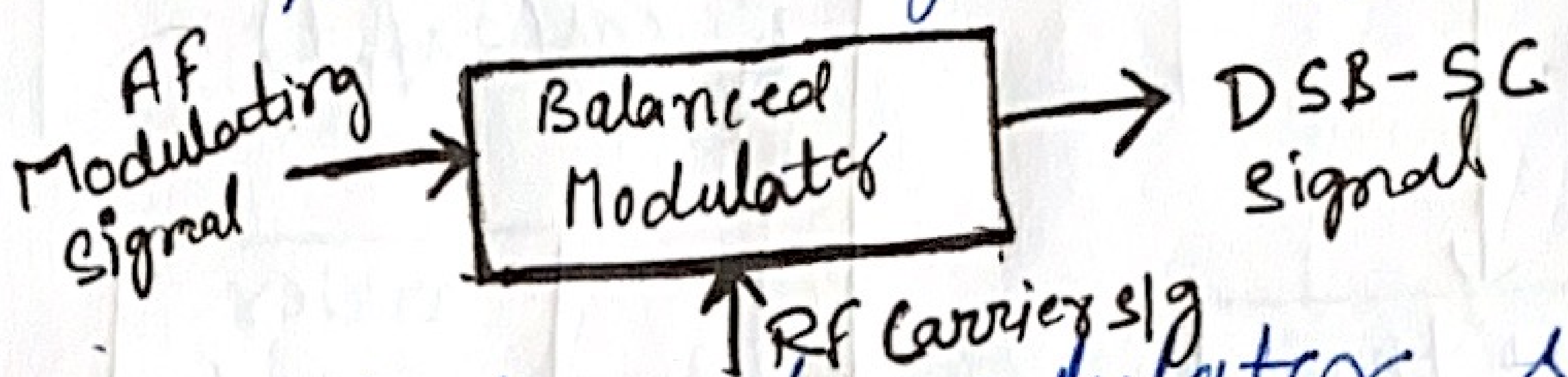
$$\boxed{V_o = K x(t) \cos\omega_c t}$$

* Diode ring modulator used for DSB-SC generation is the another version of linear modulator.

1.1 Suppression of the Carrier (Balanced Modulator)

Basic concept

The balanced modulators are used to suppress the unwanted carrier in an AM wave. The carrier and modulating signals are applied to the inputs of the balanced modular and we get the DSB signal with suppressed carrier at the output of the balanced modulator. Thus the o/p consists of the upper and lower side bands only.



Principle of operation

The principle of operation of a balanced modulator states that if two signals at different frequencies are passed through a non-linear resistance then at the output, we get an AM signal with suppressed carrier. The device having a non-linear resistance can be a diode or a JFET or even a bipolar transister.

Types of Balanced Modular

The suppression of the carrier can be done using the following two balanced Modulators:

1. Using the diode ring modulator or lattice modulator
2. Using the FET balanced modulator.

AF → 20Hz – 20KHz
RF → 30Hz – 300GHz

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1.2 Balanced Modulator using AM Modulators

Block Diagram

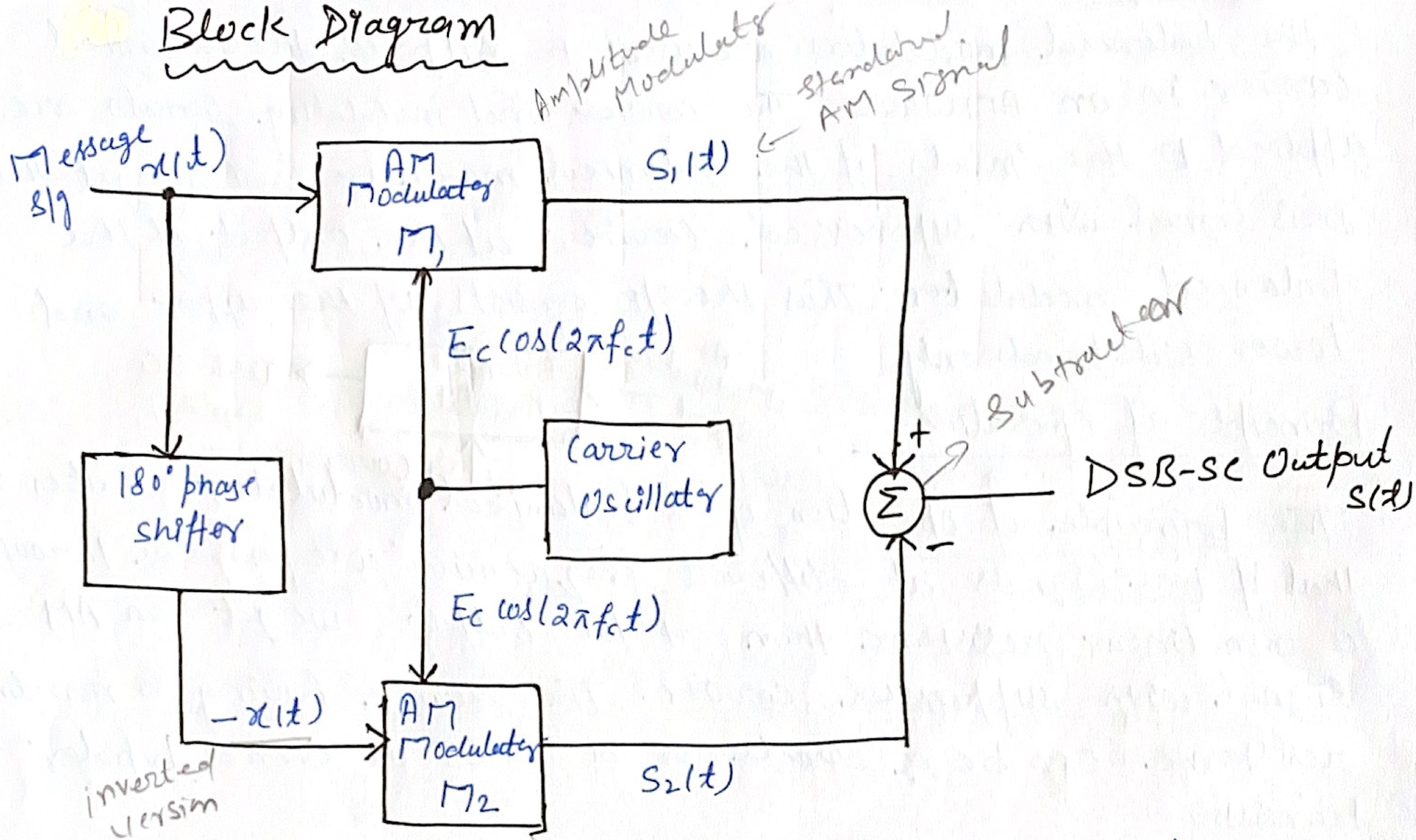


Fig. Balanced Modulator for DSB-SC generation

→ Block diagram of a balanced modulator (BM) has been shown in fig. It consists of two standards amplitude modulators arranged in the balanced configuration, so as to suppress the carrier completely.

Working

→ The carrier signal $c(t)$ is connected to both the AM Modulators M_1 and M_2.

→ The message signal $x(t)$ is applied as it is to M_1 and its inverted version $-x(t)$ is applied to M_2.

→ At the outputs of modulators M_1 and M_2 we get the standard AM signals $s_1(t)$ and $s_2(t)$ as under:-

Output of M_1 : $s_1(t) = E_c[1 + m x(t)] \cos(2\pi f_c t)$ — (1)

Output of M_2 : $s_2(t) = E_c[1 - m x(t)] \cos(2\pi f_c t)$ — (2)

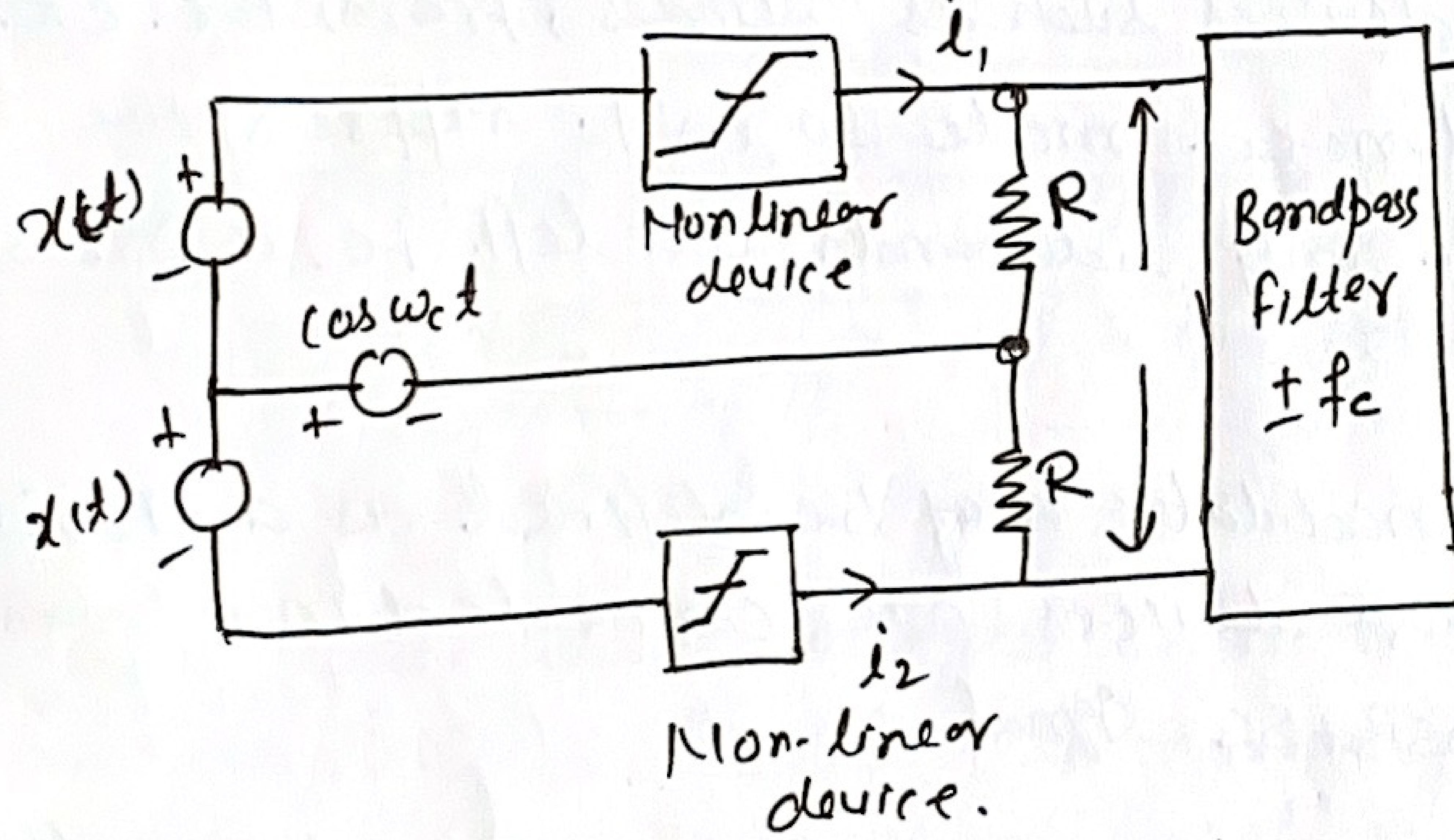
→ These are then applied to subtractor and the subtractor produce the desired DSB-SC signal as under.

$$\text{Subtractor O/P} = s_1(t) - s_2(t)$$
$$= E_c[1 + m\,x(t)]\cos(2\pi f_c t) - E_c[1 - m\,x(t)]\cos(2\pi f_c t)$$
$$= E_c\cos(2\pi f_c t)\,[\cancel{1} + m\,x(t) - \cancel{1} + m\,x(t)]$$
$$= 2m\,E_c\,x(t)\cos(2\pi f_c t) \quad \text{— (3)}$$

→ The R.H.S of this expression consists of product of $x(t)$ and $c(t) = E_c\cos(2\pi f_c t)$.
Hence, it represents a DSB-SC signal.

→ $\boxed{\text{Therefore, subtractor O/P} = \text{DSB-SC signal} = 2m\,E_c\,x(t)\cos(2\pi f_c t)}$

1.3 Modulation Using Non-linear device (Balanced Modulator)



$V_o = K\,x(t)\cos w_c t$

Fig 2 Non-linear char. of a device.

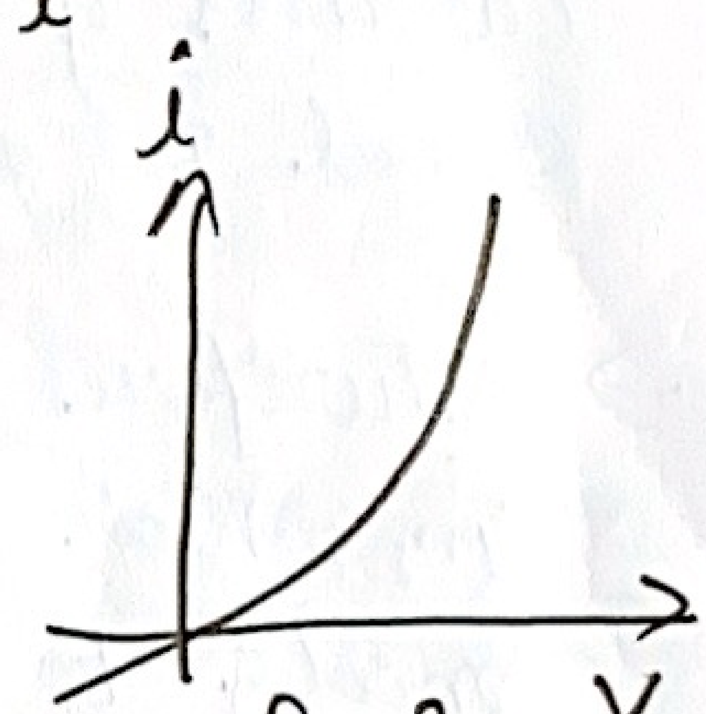


Fig 2

Fig 1 DSB-SC modulator using non-linear device.

→ The modulation can also be achieved by using the non-linear devices.

→ Fig 2 shows the non-linear device characteristics.

→ A semiconductor diode is a good example of a non-linear device.

→ The relation b/w voltage (V) and current (I) is given by

$$I = aV + bV^2, \quad \text{where } a, \text{ and } b \text{ are constants.}$$

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The balanced modulator using Diodes

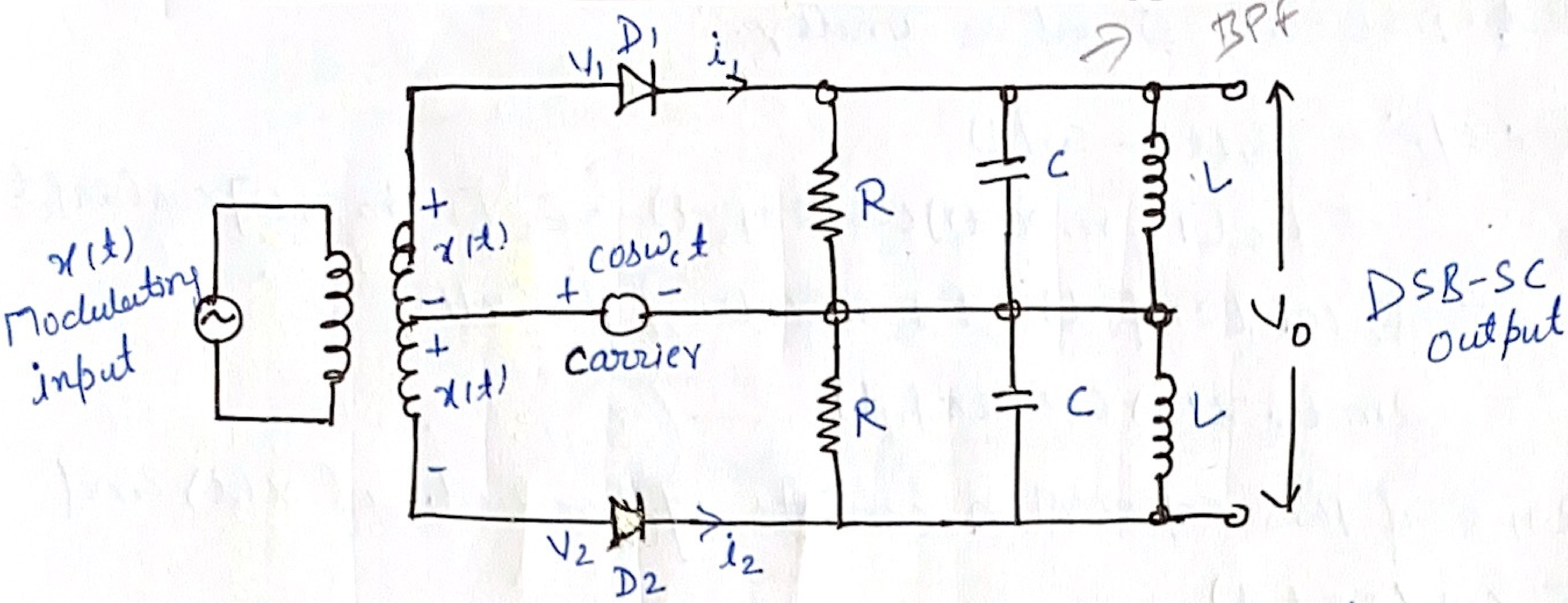

Fig Balanced modulator using diodes

→ We know that, a non-linear device may be used to produce Amplitude Modulation. i.e one carrier and two side bands.

→ However, a DSB-SC signal contains only two sidebands.

→ Thus, if two non-linear devices such as diodes, transistors etc. are connected in a balanced mode so as to suppress the carriers of each other, then only side bands are left. i.e, a DSB-SC signal is generated.

→ Therefore, a balance modulator may be defined as a circuit in which two non-linear devices are connected in a balanced mode to produce a DSB-SC signal.

→ In the fig, the modulating signal $x(t)$ is supplied equally with 180° phase reversal at the inputs of both the diodes through the input center tapped transformer.

→ The carrier is applied to the center tap of the secondary.

→ I/p voltage to D_1, is given by

$$V_1 = \cos\omega_c t + x(t) \quad \text{--- (1)}$$

I/p voltage to D_2, is given by

$$V_2 = \cos\omega_c t - x(t) \quad \text{--- (2)}$$

→ The parallel RLC circuits on the output side form the tuned BPF.

Analysis

→ The diode current i_1 and i_2 are given by,

$$i_1 = av_1 + bv_1^2$$

$$i_1 = a[x(t) + \cos\omega_c t] + b[x(t) + \cos\omega_c t]^2$$

$$i_1 = ax(t) + a\cos\omega_c t + bx^2(t) + 2bx(t)\cos\omega_c t + b\cos^2\omega_c t \quad \text{--- (3)}$$

Similarly,

$$i_2 = av_2 + bv_2^2$$

$$i_2 = a[\cos\omega_c t - x(t)] + b[\cos\omega_c t - x(t)]^2$$

$$i_2 = a\cos\omega_c t - ax(t) + bx^2(t) - 2bx(t)\cos\omega_c t + b\cos^2\omega_c t \quad \text{--- (4)}$$

→ The output voltage is given by,

$$V_0 = i_1 R - i_2 R$$

→ Substituting the expression for i_1 and i_2 from equations (3) and (4) we get.

$$V_0 = R[2ax(t)] (+ 4bx(t)\cos\omega_c t]$$

$$V_0 = \underbrace{2aRx(t)}_{\text{modulating s/g}} + \underbrace{4bRx(t)\cos\omega_c t}_{\text{DSB-SC signal}}$$

→ Hence, the output voltage contains a modulating signal term and the DSB-SC signal. The modulating s/g term is eliminated and the second term is allowed to pass through to the o/p by the LC band pass filter section.

$$\text{Final output} = 4bRx(t)\cos\omega_c t$$
$$= Kx(t)\cos\omega_c t$$

→ Thus, the diode balanced modulator produces the DSB-SC signal at it output.

Do your own

→ Ring Modulator

→ Balanced Modulator using FETs.

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1- Coherent Detection of DSB-SC Waves :-

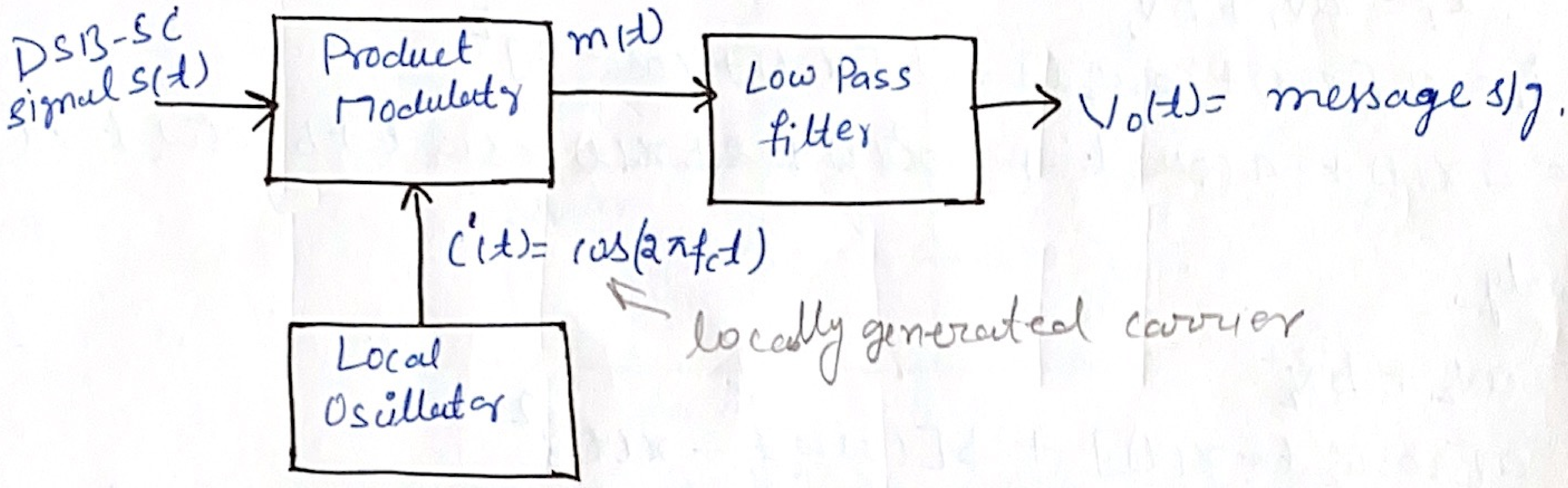

Fig. Coherent detection of DSB-SC modulated wave.

→ The DSB-SC wave s(t) is applied to a product modulator in which it is multiplied with the locally generated carrier $\cos(2\pi f_c t)$.

→ We assume that this locally generated carrier is exactly synchronized in both freq. and phase, with the original carrier wave c(t) used to generate the DSB-SC wave.

→ This method of detection is therefore called as coherent detection or synchronous detection.

→ The output of the product modulator is applied to the low pass filter (LPF) which eliminates all the unwanted freq. components and produces the message signal.

Analysis of Coherent Detection :-

→ Let the output of the local oscillator be given by,

$$c'(t) = \cos(2\pi f_c t + \phi)$$

Thus its amplitude is 1 (unity), freq. is f_c and the phase difference is arbitrary equal to ϕ. This phase difference has been measured w.r.t the original carrier c(t) at the DSB-SC generator.

Therefore, the output of the product modulator is given by,

$$m(t) = s(t) \cdot c'(t)$$

49575
49612

$s(t)$ = DSB-SC input $= x(t) \cdot E_c \cos(2\pi f_c t)$

$c'(t)$ = Local carrier $= \cos(2\pi f_c t + \phi)$

$m(t) = x(t) \cdot E_c \cos(2\pi f_c t) \cos(2\pi f_c t + \phi) \rightarrow \boxed{m(t) = s(t) \cdot c'(t)}$

$m(t) = x(t) \cdot E_c \cos(2\pi f_c t + \phi) \cos(2\pi f_c t)$

$\boxed{\text{But, } \cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]}$

Hence $\cos(2\pi f_c t + \phi) \cos(2\pi f_c t) = \frac{1}{2}[\cos(2\pi f_c t + \phi + 2\pi f_c t) + \cos\phi]$

$= \frac{1}{2}[\cos(4\pi f_c t + \phi) + \cos\phi]$

Thus, $m(t) = \frac{1}{2} x(t) E_c [\cos(4\pi f_c t + \phi) + \cos\phi]$

→ carrier → modulated s/g

$$m(t) = \underbrace{\frac{1}{2} E_c \cos\phi \, x(t)}_{\text{Scaled version of message s/g, } x(t)} + \underbrace{\frac{1}{2} x(t) E_c \cos(4\pi f_c t + \phi)}_{\text{Unwanted term}} \quad \text{—— (1)}$$

O/P of product modulator

→ ~~The first one~~

→ Eq (1) shows that the O/P of product modulator i.e $m(t)$ consists of two terms. The first one represents the message s/g $x(t)$

6.75
36.50

☆ Demodulation of DSB-SC by Switching (Chopper) Demodulator

1.1

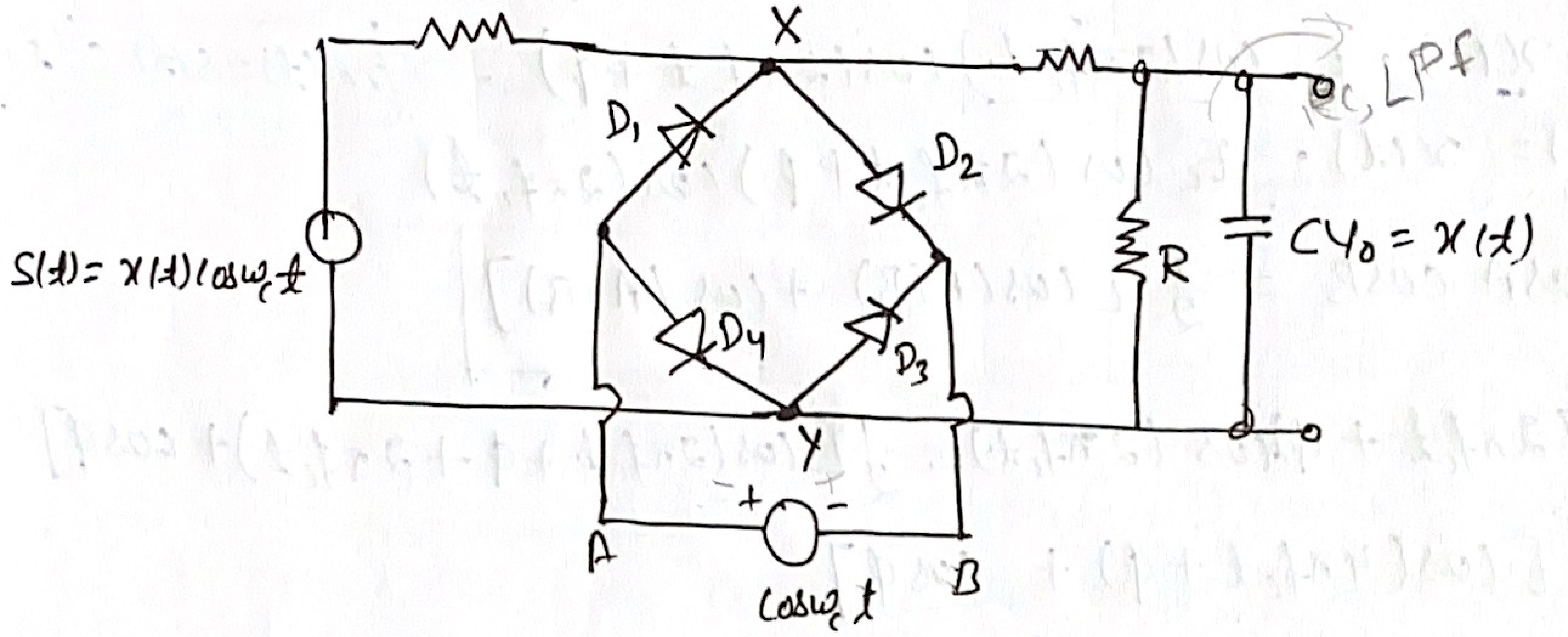

Fig.1 Switching demodulator for DSB-SC

→ fig. shows the switching (chopper) demodulator for the DSB-SC wave.

→ The diodes act as switches.

→ The DSB-SC wave $s(t)$ is applied at the input whereas the locally generated synchronous carrier $\cos\omega_c t$ is applied b/w the diodes.

→ An RC, LPF is connected across the outputs terminals.

→ In the, +ve half cycle of carrier $\cos\omega_c t$, point A is +ve w.r.t. B. So, all the diodes are forward biased and act as closed switches. Point X and Y are connected to each other and $V_{XY} = 0$.

→ In the, −ve half cycle of $\cos\omega_c t$, point B is +ve w.r.t A. So, all the diodes are reverse biased and act as open switches. Hence, the DSB-SC input $s(t)$ is allowed to pass through to the output.

→ Thus, only one half cycle (negative) of $s(t)$ is allowed while the +ve half is suppressed.

→ Therefore, effectively the rectification of $s(t)$ is taking place. **Rectification** process is equivalent to multiplication.

$$V_0(t) = s(t) \times \cos\omega_c t$$

$$V_0(t) = x(t)\cos\omega_c t \times \cos\omega_c t$$

$$V_0(t) = x(t)\cos^2\omega_c t$$

$$V_0(t) = x(t)\left[\frac{1+\cos 2\omega_c t}{2}\right]$$

$$\boxed{V_0(t) = \frac{x(t)}{2} + \frac{1}{2}\cos(2\omega_c t)x(t)}$$

→ Taking the fourier transform of this expression, we get the spectrum of V_0 as under:

$$F[V_0(t)] = V_0(f)$$

$$= \frac{1}{2}X(f) + \frac{1}{2}[\delta(f-2f_c) + \delta(f+2f_c)]X(f)$$

where, $X(f)$ = spectrum of $x(t)$

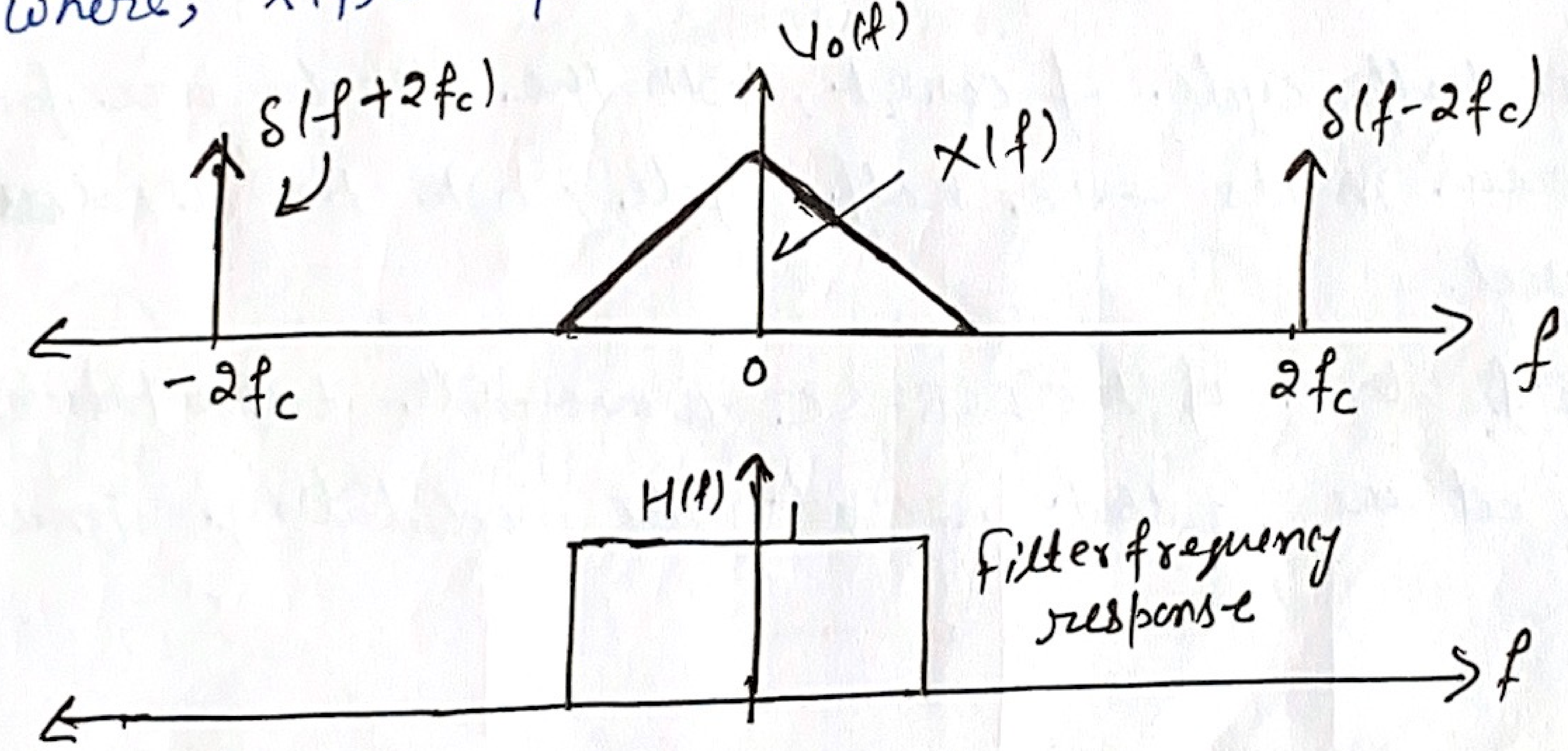

Fig. 2 spectrum of $V_0(t)$

Fig 2 shows that the demodulator o/p consists of spectrum of $X(t)$ and second harmonic of carrier ($2f_c$). We use a RC low pass filter to pass only $X(f)$ through diode to the output. Hence, the demodulator output contains only the modulating signal $x(t)$.

1.2 DSB-SC Demodulator using Non-linear Devices

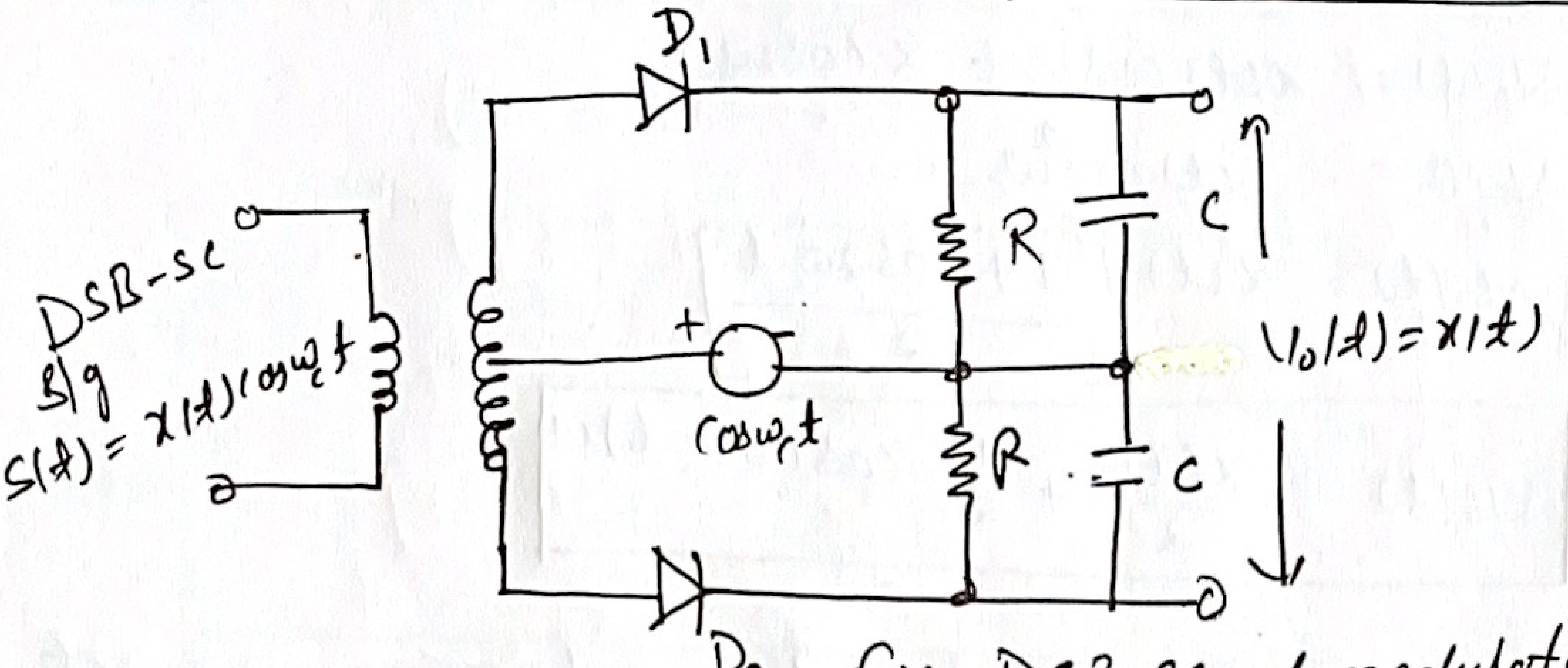

fig DSB-SC demodulator using non-linear devices.

→ The DSB-SC signal s(t) is applied at the input of the center tap transformer.

→ A locally generated synchronised carrier $\cos\omega_c t$ is applied to both the diodes via the two secondary winding of the i/p transformer.

→ In the +ve half cycle of $\cos\omega_c t$, both the diodes are forward biased whereas in its −ve half cycle, both the diodes are reverse biased.

→ Hence, rectification of the DSB-SC wave will take place.

→ Therefore, at the output, we get the modulating signal x(t) back.